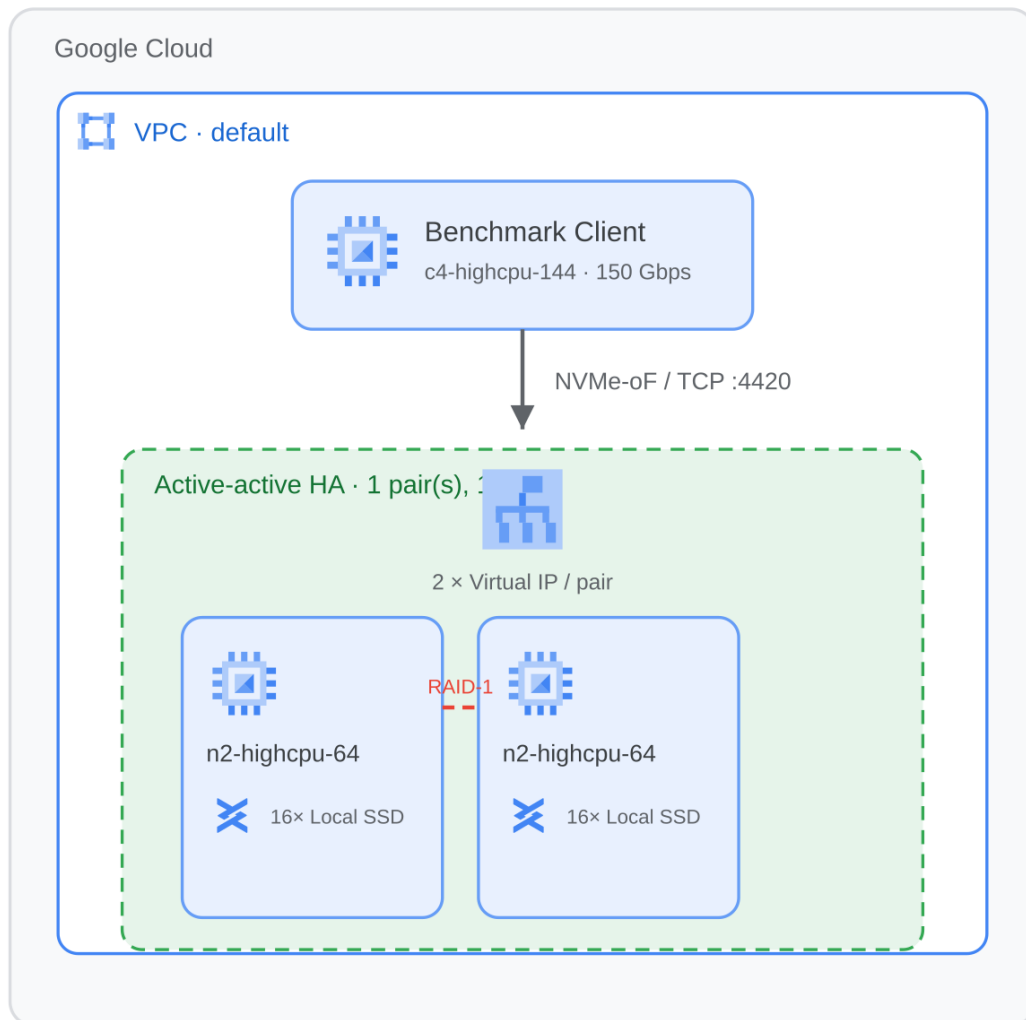


MayaScale Block — MLPerf Storage v3.0 (RetinaNet + KVCache)

Submitter: ZettaLane Systems · **Division:** Closed · **System:** mayascale-gcp_c4hc144_n2hc64_1c_2s

architecture

MayaScale — NVMe-oF Block on Google Cloud



MayaScale Storage Cluster (2 nodes, 1 active-active HA pair)

- 2 × GCP n2-highcpu-64 (Intel Xeon, 64 vCPU / 2×16 cores, 62 GB RAM)

- Rocky Linux 9.7 · 75 Gbps networking · NVMe-oF/TCP target
- 16 × Local SSD (375 GiB, NVMe TLC) per node (32 total, 11.72 TiB raw → 5.86 TiB usable after mirror)
- Cross-node RAID-1 mirror (nvmetmirror) → 16 mirrored NVMe-oF volumes exported

Benchmark Client

- 1 × GCP c4-highcpu-144 (Intel Xeon 6985P-C (Granite Rapids), 144 vCPU, 283 GB RAM)
- Ubuntu 24.04 · 150 Gbps networking
- NVMe-oF/TCP initiator → mdadm RAID0 over mirrored volumes (RAID10) → XFS

Results

- RetinaNet (na=72, 72× B200): mean Accelerator Utilization 86.68%
- KVCache (llama3.1-8b, stress): 526 tok/s, 8.46 GB/s read

Availability

MayaNAS and MayaScale can be deployed from the Google Cloud Marketplace portal or as Terraform modules (<https://github.com/zettalane-systems/open-lustre-cloud>). Contact: info@zettalane.com.